

Unit 3

Exploratory Data Analytics: Descriptive Statistics, Mean, Standard Deviation, Skewness and Kurtosis, Box Plots, Pivot Table, Heat Map, Correlation Statistics.

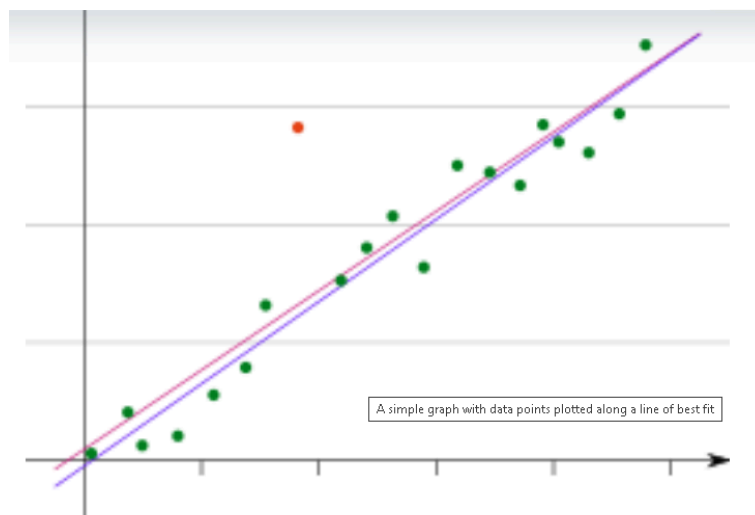
Exploratory Data Analytics: -

Exploratory Data Analysis (EDA) is an approach to analyze the data using visual techniques. It is used to discover trends, patterns, or to check assumptions with the help of statistical summary and graphical representations.

In data analytics, exploratory data analysis is how we describe the practice of investigating a dataset and summarizing its main features. It is a form of descriptive analytics. EDA aims to spot patterns and trends, to identify anomalies, and to test early hypotheses. Although exploratory data analysis can be carried out at various stages of the data analytics process, it is usually conducted before a firm hypothesis or end goal is defined.

In general, EDA focuses on understanding the characteristics of a dataset before deciding what we want to do with that dataset.

Exploratory data analytics often uses visual techniques, such as graphs, plots, and other visualizations. This is because our natural pattern-detecting abilities make it much easier to spot trends and anomalies when they're represented visually. As a simple example, outliers (or data points that skew a trend) stand out much more immediately on a scatter graph than they do in columns on a spreadsheet.



Source: Indoor-Fanatiker derivative work: Indoor-Fanatiker. GFDL via [Wikimedia Commons](#)

In the image, the outlier (in red) is immediately clear. Even if you're new to data analytics, this approach will seem familiar. If you ever plotted a graph in math or science at school in order to infer information about a dataset, then you've carried out a basic EDA.

The American mathematician John Tukey formally introduced the concept of exploratory data analysis in 1961. The idea of summarizing a dataset's key characteristics coincided with the development of early programming languages such as R and S. This was also a time when

scientists and engineers were working on new data-driven problems related to early computing. Since then, EDA has been widely adopted as a core tenet of data analytics and data science more generally. It is now considered a common—indeed, indispensable—part of the data analytics process.

Why is exploratory data analysis important in data science?

The main purpose of EDA is to help look at data before making any assumptions. It can help identify obvious errors, as well as better understand patterns within the data, detect outliers or anomalous events, find interesting relations among the variables.

Data scientists can use exploratory analysis to ensure the results they produce are valid and applicable to any desired business outcomes and goals. EDA also helps stakeholders by confirming they are asking the right questions. EDA can help answer questions about standard deviations, categorical variables, and confidence intervals. Once EDA is complete and insights are drawn, its features can then be used for more sophisticated data analysis or modeling, including machine learning.

Exploratory data analysis tools

Specific statistical functions and techniques you can perform with EDA tools include:

Clustering and dimension reduction techniques, which help create graphical displays of high-dimensional data containing many variables. Univariate visualization of each field in the raw dataset, with summary statistics. Bivariate visualizations and summary statistics that allow you to assess the relationship between each variable in the dataset and the target variable you're looking at. Multivariate visualizations, for mapping and understanding interactions between different fields in the data.

K-means Clustering is a clustering method in unsupervised learning where data points are assigned into K groups, i.e. the number of clusters, based on the distance from each group's centroid. The data points closest to a particular centroid will be clustered under the same category. K-means Clustering is commonly used in market segmentation, pattern recognition, and image compression.

Predictive models, such as linear regression, use statistics and data to predict outcomes.

Types of exploratory data analysis

There are four primary types of EDA:

Univariate non-graphical. This is simplest form of data analysis, where the data being analyzed consists of just one variable. Since it's a single variable, it doesn't deal with causes or relationships. The main purpose of univariate analysis is to describe the data and find patterns that exist within it.

Univariate graphical. Non-graphical methods don't provide a full picture of the data. Graphical methods are therefore required. Common types of univariate graphics include:

Stem-and-leaf plots, which show all data values and the shape of the distribution.

Histograms, a bar plot in which each bar represents the frequency (count) or proportion (count/total count) of cases for a range of values.

Box plots, which graphically depict the five-number summary of minimum, first quartile, median, third quartile, and maximum.

Multivariate nongraphical: Multivariate data arises from more than one variable. Multivariate non-graphical EDA techniques generally show the relationship between two or more variables of the data through cross-tabulation or statistics.

Multivariate graphical: Multivariate data uses graphics to display relationships between two or more sets of data. The most used graphic is a grouped bar plot or bar chart with each group representing one level of one of the variables and each bar within a group representing the levels of the other variable.

Other common types of multivariate graphics include:

Scatter plot, which is used to plot data points on a horizontal and a vertical axis to show how much one variable is affected by another.

Multivariate chart, which is a graphical representation of the relationships between factors and a response.

Run chart, which is a line graph of data plotted over time.

Bubble chart, which is a data visualization that displays multiple circles (bubbles) in a two-dimensional plot.

Heat map, which is a graphical representation of data where values are depicted by color.

Exploratory Data Analysis Tools

Some of the most common data science tools used to create an EDA include:

Python: An interpreted, object-oriented programming language with dynamic semantics. Its high-level, built-in data structures, combined with dynamic typing and dynamic binding, make it very attractive for rapid application development, as well as for use as a scripting or glue language to connect existing components together. Python and EDA can be used together to identify missing values in a data set, which is important so you can decide how to handle missing values for machine learning.

R: An open-source programming language and free software environment for statistical computing and graphics supported by the R Foundation for Statistical Computing. The R language is widely used among statisticians in data science in developing statistical observations and data analysis.

- The term “descriptive statistics” refers to the analysis, summary, and presentation of findings related to a data set derived from a sample or entire population.
- Descriptive statistics comprises three main categories – Frequency Distribution, Measures of Central Tendency, and Measures of Variability.
- Descriptive statistics helps facilitate data visualization. It allows for data to be presented in a meaningful and understandable way, which, in turn, allows for a simplified interpretation of the data set in question.

What Are Descriptive Statistics?

Descriptive statistics are brief informational coefficients that summarize a given data set, which can be either a representation of the entire population or a sample of a population. Descriptive statistics are broken down into measures of central tendency and measures of variability (spread). Measures of central tendency include the mean, median, and mode, while measures of variability include standard deviation, variance, minimum and maximum variables, kurtosis, and skewness.

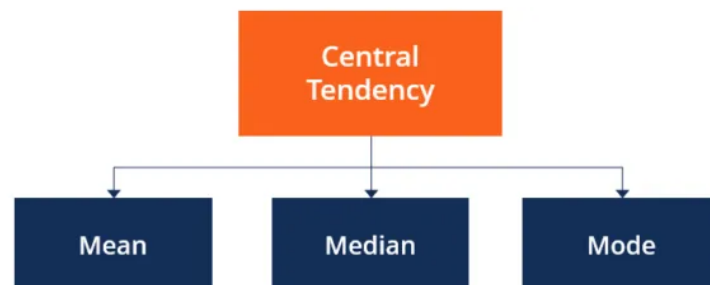
Understanding the Different Types of Descriptive Statistics

Frequency Distribution: -Used for both quantitative and qualitative data, frequency distribution depicts the frequency or count of the different outcomes in a data set or sample. The frequency distribution is normally presented in a table or a graph. Each entry in the table or graph is accompanied by the count or frequency of the values’ occurrences in an interval, range, or specific group.

Frequency distribution is basically a presentation or summary of grouped data categorized based on mutually exclusive classes and the number of occurrences in each respective class. It allows for a more structured and organized way to present raw data.

Common charts and graphs used in frequency distribution presentation and visualization include bar charts, histograms, pie charts, and line charts.

Central Tendency



Measures of Central Tendency Central tendency refers to a dataset’s descriptive summary using a single value reflecting the center of the data distribution. Measures of central tendency are also known as measures of central location. The mean, median, and mode are the measures of central tendency.

The mean, considered the most popular measure of central tendency, is the average or most common value in a data set. The median refers to the middle score for a data set in ascending order. The mode refers to the score or value that is most frequent in a data set.

Variability A measure of variability is a summary statistic reflecting the degree of dispersion in a sample. The measures of variability determine how far apart the data points appear to fall from the center. Dispersion, spread, and variability all refer to and denote the range and width of the distribution of values in a data set. The range, standard deviation, and variance are used, respectively, to depict different components and aspects of the spread.

The range depicts the degree of dispersion or an ideal of the distance between the highest and lowest values within a data set. The standard deviation is used to determine the average variance in a set of data and provide an insight into the distance or difference between a value in a data set and the mean value of the same data set. The variance reflects the degree of the spread and is essentially an average of the squared deviations.

For a deeper understanding of different foundational statistics concepts and tools, check out CFI's Statistics Fundamentals course!

MEAN, MEDIAN, MODE

The mean median mode are measurements of central tendency. In other words, it tells you where the “middle” of a data set it. Each of these statistics defines the middle differently:

1. The **mean** is the average of a data set.
2. The **mode** is the most common number in a data set.
3. The **median** is the middle of the set of numbers.

Of the three, the mean is the only one that requires a formula. I like to think of it in the other dictionary sense of the word (as in, it's “mean” as opposed to nice!). That's because, compared to the other two, it's not as easy to work with because of the formula.

Mean vs Median

Both are measures of where the center of a data set lies (called “Central Tendency” in stats), but they are usually *different* numbers. For example, take this list of numbers: 10, 10, 20, 40, 70.

- The mean (informally, the “average”) is found by adding all of the numbers together and dividing by the number of items in the set: $10 + 10 + 20 + 40 + 70 / 5 = 30$.
- The median is found by ordering the set from lowest to highest and finding the exact middle. The median is just the middle number: 20.

Sometimes the two will be the same number. For example, the data set 1, 2, 4, 6, 7 has a mean of $1 + 2 + 4 + 6 + 7 / 5 = 4$ and a median (a middle) of 4.

Mean vs Average: What's the Difference?

When you first started out in mathematics, you were probably taught that an average was a “middling” amount for a set of numbers. You added up the numbers, divided by the number of items you can and *voila!* you get the average. For example, the average of 10, 6 and 20 is:

$$10 + 6 + 20 = 36 / 3 = 12.$$

Then you started studying statistics and all of a sudden the “average” is now called the mean. *What happened?* The answer is that they have the same meaning (they are synonyms).

However, there is a caveat. *Technically*, the word *mean* is short for the arithmetic mean. We use different words in stats, because there are multiple different types of means, and they all do different things.

Specific “Means” commonly used in Stats

You’ll probably come across these in an elementary stats class. They have very narrow meanings:

- Mean of the sampling distribution: the center of a probability distribution, especially with respect to the Central Limit Theorem. It’s an average (of sorts) of a set of distributions.
- Sample mean: the average value in a sample.
- Population mean: the average value in a population.

Types of Mean: -

- Weighted Mean

These are fairly common in statistics, especially when studying populations. Instead of each data point contributing equally to the final average, some data points contribute more than others. If all the weights are equal, then this will equal the arithmetic mean. There are certain circumstances when this can give incorrect information, as shown by Simpson’s Paradox.

- Harmonic Mean

$$H(x_1, \dots, x_n) = \frac{n}{\frac{1}{x_1} + \dots + \frac{1}{x_n}}$$

The harmonic formula.

To find it:

- Add the reciprocals of the numbers in the set. To find a reciprocal, flip the fraction so that the numerator becomes the denominator and the denominator becomes the numerator. For example, the reciprocal of 6/1 is 1/6.
- Divide the answer by the number of items in the set.
- Take the reciprocal of the result.

The harmonic mean is used quite a lot in physics. In some cases involving rates and ratios it gives a better average than the arithmetic mean. In addition, you’ll also find uses in geometry, finance and computer science.

- Geometric Mean

$$G(x_1, \dots, x_n) = \sqrt[n]{x_1 \cdots x_n}$$

This type has very narrow and specific uses in finance, social sciences and technology. For example, let's say you own stocks that earn 5% the first year, 20% the second year, and 10% the third year. If you want to know the average rate of return, you can't use the arithmetic average. Why? Because when you are finding rates of return you are *multiplying*, not adding. For example, the first year at a 5% earnings rate you will multiply by 1.05.

- Arithmetic-Geometric Mean

This is used mostly in calculus and in machine computation (i.e. as the basis for many computer calculations). It's related to the perimeter of an ellipse and when it was first developed by Gauss, it was used to calculate planetary orbits. The arithmetic-geometric is (perhaps not surprisingly) a blend of the arithmetic and geometric averages. The math is quite complicated but you can find a relatively simple explanation of the math [here](#).

- Root-Mean Square

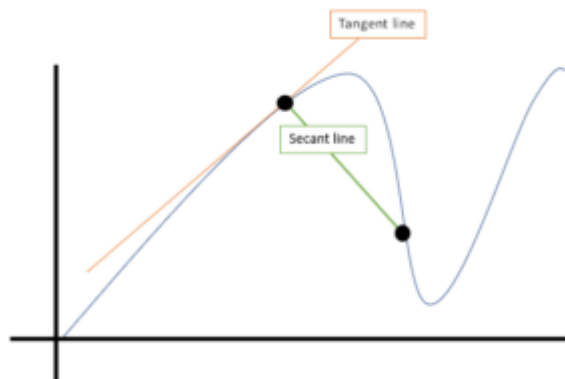
It is very useful in fields that study sine waves, like electrical engineering. This particular type is also called the quadratic average. See: Quadratic Mean / Root Mean Square.

- Heronian Mean

Used in geometry to find the volume of a pyramidal frustum. A pyramidal frustum is basically a pyramid with the tip sliced off.

- Graphic Mean

Another name for the slope of the secant line: the equivalent of the average rate of change between two points.



2. What is the Mode?

The mode is the most **common number** in a set. For example, the mode in this set of numbers is 21:

21, 21, 21, 23, 24, 26, 26, 28, 29, 30, 31, 33

3. What is the Median?

The median is the **middle number** in a data set. To find the median, list your data points in ascending order and then find the middle number. The middle number in this set is 28 as there are 4 numbers below it and 4 numbers above:

23, 24, 26, 26, 28, 29, 30, 31, 33

Note: If you have an even set of numbers, average the middle two to find the median. For example, the median of this set of numbers is 28.5 ($28 + 29 / 2$):

23, 24, 26, 26, 28, 29, 30, 31, 33, 34

How to find the mean median mode by hand: Steps

How to find the mean median mode: MODE

- **Step 1:** *Put the numbers in order so that you can clearly see patterns.*
For example, let's say we have 2, 19, 44, 44, 44, 51, 56, 78, 86, 99, 99.
The **mode** is the number that appears the most often. In this case: **44**, which appears three times.

How to find the mean median mode: MEAN

- **Step 2:** *Add the numbers up to get a total.*
Example: $2 + 19 + 44 + 44 + 44 + 51 + 56 + 78 + 86 + 99 + 99 = 622$. Set this number aside for a moment.
- **Step 3:** *Count the amount of numbers in the series.*
In our example (2, 19, 44, 44, 44, 51, 56, 78, 86, 99, 99), we have **11** numbers.
- **Step 4:** *Divide the number you found in step 2 by the number you found in step 3.*
In our example: $622 / 11 = 56.5454545$. This is the **mean**, sometimes called the **average**.

How to find the mean median mode: MEDIAN

If you had an odd number in step 3, go to step 5. If you had an even number, go to step 6.

- **Step 5:** Find the number in the middle of the list.
This is the **median**. ~~2, 19, 44, 44, 44, 51, 56, 78, 86, 99, 99~~.
- **Step 6:** Find the middle two numbers.
For example, ~~1, 2, 5, 6, 7, 8, 12, 15, 16, 17~~. The **median** is the number that comes in the middle of those middle two numbers (7 and 8), so that number would be 7.5 in this case. In order to do this mathematically, add the two numbers together and divide by 2.

Tip: You can have more than one mode. For example, the three modes of 1, 1, 5, 5, 6, 6 are 1, 5, and 6.

Let Us Understand the Above 5 Statistics Formulas with Examples:

- **Mean:** The arithmetical mean is the sum of a set of **numbers** separated by the number of numbers in the collection, or simply the mean or the **average**.
- **Median:** In a sorted, ascending or descending, list of numbers, the median is the middle number and may be more representative of that data set than the average.

(Image to be added soon)

- **Mode:** The mode is the value that most frequently appears in a data value set.

(Image to be added soon)

- **Standard Deviation:** A calculation of the amount of variance or dispersion of a set of values is the standard deviation.

(Image to be added soon)

- **Variance:** The expectation of the **square** deviation of a random **variable** from its mean is variance.

Now let us look at the formula of statistics that can be used while solving the problems.

The Mean and Mode

The **sample mean** is the **average** and is computed as the sum of all the observed outcomes from the sample divided by the total number of events. We use **x** as the symbol for the sample mean. In math terms,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x$$

where **n** is the **sample size** and the **x** correspond to the **observed valued**.

Example: Suppose you randomly sampled six acres in the Desolation Wilderness for a non-indigenous weed and came up with the following counts of this weed in this region:

34, 43, 81, 106, 106 and 115

We compute the sample mean by adding and dividing by the number of samples, 6.

$$\frac{34 + 43 + 81 + 106 + 106 + 115}{6} = 80.83$$

We can say that the sample mean of non-indigenous weed is 80.83.

The *mode* of a set of data is the number with the highest frequency. In the above example 106 is the mode, since it occurs twice and the rest of the outcomes occur only once.

The *population mean* is the *average* of the *entire population* and is usually impossible to compute. We use the Greek letter μ for the population mean.

Median, and Trimmed Mean

One problem with using the mean, is that it often does not depict the typical outcome. If there is one outcome that is very far from the rest of the data, then the mean will be strongly affected by this outcome. Such an outcome is called an *outlier*. An alternative measure is the median. The *median* is the *middle score*. If we have an even number of events we take the average of the two middles. The median is better for describing the typical value. It is often used for income and home prices.

Example

Suppose you randomly selected 10 house prices in the South Lake Tahoe area. You are interested in the typical house price. In \$100,000 the prices were

2.7, 2.9, 3.1, 3.4, 3.7, 4.1, 4.3, 4.7, 4.7, 40.8

If we computed the mean, we would say that the average house price is 744,000. Although this number is true, it does not reflect the price for available housing in South Lake Tahoe. A closer look at the data shows that the house valued at 40.8 x \$100,000 = \$4.08 million skews the data. Instead, we use the median. Since there is an even number of outcomes, we take the average of the middle two

$$\frac{3.7 + 4.1}{2} = 3.9$$

The median house price is \$390,000. This better reflects what house shoppers should expect to spend.

There is an alternative value that also is resistant to outliers. This is called the *trimmed mean* which is the mean after getting rid of the outliers or 5% on the top and 5% on the bottom. We can also use the trimmed mean if we are concerned with outliers skewing the data, however the median is used more often since more people understand it.

Example:

At a ski rental shop data was collected on the number of rentals on each of ten consecutive Saturdays:

44, 50, 38, 96, 42, 47, 40, 39, 46, 50.

To find the sample mean, add them and divide by 10:

$$\frac{44 + 50 + 38 + 96 + 42 + 47 + 40 + 39 + 46 + 50}{10} = 49.2$$

Notice that the mean value is not a value of the sample.

To find the median, first sort the data:

38, 39, 40, 42, 44, 46, 47, 50, 50, 96

Notice that there are two middle numbers 44 and 46. To find the median we take the average of the two.

$$\text{Median} = \frac{44 + 46}{2} = 45$$

Notice also that the mean is larger than all but three of the data points. The mean is influenced by outliers while the median is robust.

Variance, Standard Deviation and Coefficient of Variation

The mean, mode, median, and trimmed mean do a nice job in telling where the centre of the data set is, but often we are interested in more. For example, a pharmaceutical engineer develops a new drug that regulates iron in the blood. Suppose she finds out that the average sugar content after taking the medication is the optimal level. This does not mean that the drug is effective. There is a possibility that half of the patients have dangerously low sugar content while the other half have dangerously high content. Instead of the drug being an effective regulator, it is a deadly poison. What the pharmacist needs is a measure of how far the data is spread apart. This is what the variance and standard deviation do. First we show the formulas for these measurements. Then we will go through the steps on how to use the formulas.

We define the *variance* to be

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

and the *standard deviation* to be

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x - \bar{x})^2}$$

Variance and Standard Deviation: Step by Step

1. Calculate the mean, $\bar{x}$.
2. Write a table that subtracts the mean from each observed value.
3. Square each of the differences.
4. Add this column.
5. Divide by $n - 1$ where n is the number of items in the sample This is the *variance*.
6. To get the *standard deviation* we take the square root of the variance.

Example

The owner of the Ches Tahoe restaurant is interested in how much people spend at the restaurant. He examines 10 randomly selected receipts for parties of four and writes down the following data.

44, 50, 38, 96, 42, 47, 40, 39, 46, 50

He calculated the mean by adding and dividing by 10 to get

$$\bar{x} = 49.2$$

Below is the table for getting the standard deviation:

x	$x - 49.2$	$(x - 49.2)^2$
44	-5.2	27.04
50	0.8	0.64
38	11.2	125.44
96	46.8	2190.24
42	-7.2	51.84
47	-2.2	4.84
40	-9.2	84.64
39	-10.2	104.04
46	-3.2	10.24
50	0.8	0.64
Total		2600.4

Now

$$\frac{2600.4}{10 - 1} = 288.7$$

Hence the variance is 289 and the standard deviation is the square root of 289 = 17. Since the standard deviation can be thought of measuring how far the data values lie from the mean, we take the mean and move one standard deviation in either direction. The mean for this example was about 49.2 and the standard deviation was 17. We have:

$$49.2 - 17 = 32.2$$

and

$$49.2 + 17 = 66.2$$

(<https://atozmath.com/example/StatsG.aspx?q=5&q1=A1>

<https://atozmath.com/example/StatsG.aspx?q=5&q1=E1>

<https://atozmath.com/example/StatsG.aspx?q=5&q1=E2>),

Box Plot: -

Box Plot: It is a type of chart that depicts a group of numerical data through their quartiles. It is a simple way to visualize the shape of our data. It makes comparing characteristics of data between categories very easy.

In this article, we are going to discuss the following topics-

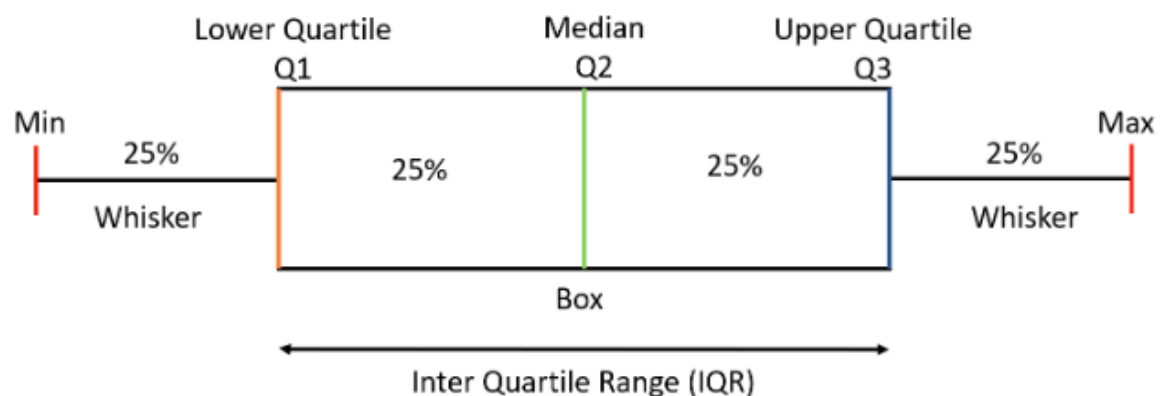
1. Understanding the components of a box plot
2. How to create a box plot
3. Uses of a Box Plot
4. How to compare box plots

Let's proceed further step by step

1) Understanding the components of a box plot

A box plot gives a five-number summary of a set of data which is-

- **Minimum** – It is the minimum value in the dataset excluding the outliers
- **First Quartile (Q1)** – 25% of the data lies below the First (lower) Quartile.
- **Median (Q2)** – It is the mid-point of the dataset. Half of the values lie below it and half above.
- **Third Quartile (Q3)** – 75% of the data lies below the Third (Upper) Quartile.
- **Maximum** – It is the maximum value in the dataset excluding the outliers.



Note: The box plot shown in the above diagram is a perfect plot with no skewness. The plots can have skewness and the median might not be at the center of the box.

The area inside the box (50% of the data) is known as the **Inter Quartile Range**. The **IQR** is calculated as –

$$\text{IQR} = Q3 - Q1$$

Outliers are the data points **below and above** the **lower and upper limit**. The lower and upper limit is calculated as –

$$\text{Lower Limit} = Q1 - 1.5 * \text{IQR}$$

$$\text{Upper Limit} = Q3 + 1.5 * \text{IQR}$$

The values below and above these limits are considered outliers and the minimum and maximum values are calculated from the points which lie under the lower and upper limit.

2) How to create a box plot

Let us take a sample data to understand how to create a box plot.

Here are the runs scored by a cricket team in a league of 12 matches –
100,120,110,150,110,140,130,170,120,220,140,110.

To draw a box plot for the given data first we need to arrange the data in ascending order and then find the minimum, first quartile, median, third quartile and the maximum.

Ascending Order -

100,110,110,110,120,120,130,140,140,150,170,220

Median (Q2) = $(120+130)/2 = 125$; Since there were even values

To find the First Quartile we take the first six values and find their median.

$$\text{Q1} = (110+110)/2 = 110$$

For the Third Quartile, we take the next six and find their median.

$$\text{Q3} = (140+150)/2 = 145$$

Note: If the total number of values is odd then we exclude the Median while calculating Q1 and Q3. Here since there were two central values we included them.

Now, we need to calculate the Inter Quartile Range.

$$\text{IQR} = Q3 - Q1 = 145 - 110 = 35$$

We can now calculate the Upper and Lower Limits to find the minimum and maximum values and also the outliers if any.

$$\text{Lower Limit} = Q1 - 1.5 * \text{IQR} = 110 - 1.5 * 35 = 57.5$$

$$\text{Upper Limit} = Q3 + 1.5 * \text{IQR} = 145 + 1.5 * 35 = 197.5$$

So the minimum and maximum between the range [57.5,197.5] for our given data are –

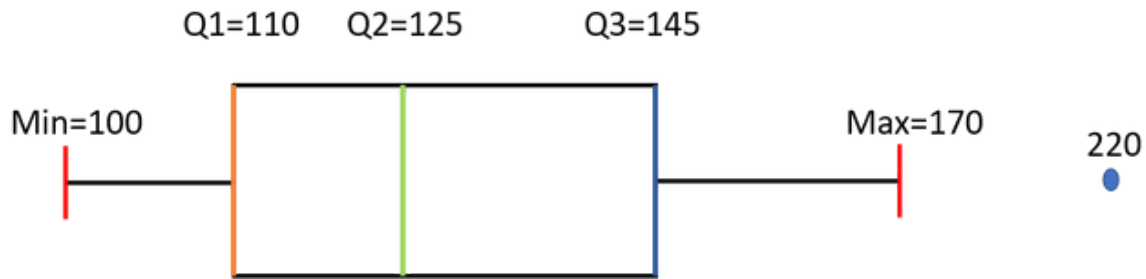
Minimum = 100

Maximum = 170

The outliers which are outside this range are –

Outliers = 220

Now we have all the information, so we can draw the box plot which is as below-

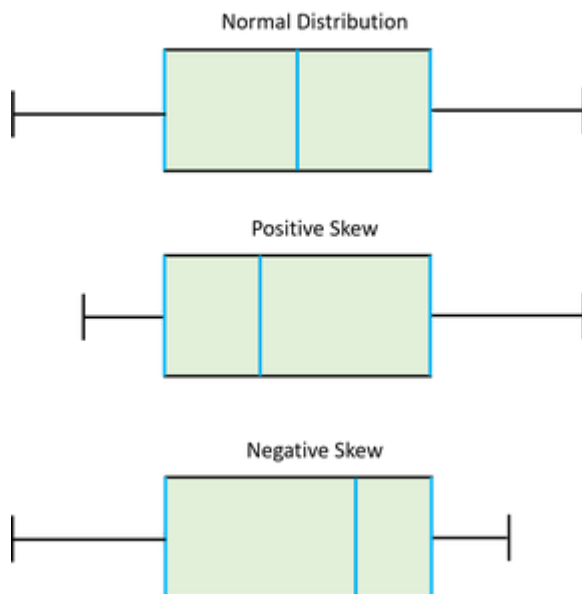


We can see from the diagram that the Median is not exactly at the centre of the box and one whisker is longer than the other. We also have one Outlier.

3) Uses of a Box Plot

- Box plots provide a visual summary of the data with which we can quickly identify the average value of the data, how dispersed the data is, whether the data is skewed or not (skewness).
- The Median gives you the average value of the data.
- Box Plots shows Skewness of the data-

- If the Median is at the **center** of the Box and the **whiskers** are almost the **same on both the ends** then the data is **Normally Distributed**.
- If the Median lies **closer to the First Quartile** and if the **whisker at the lower end is shorter** (as in the above example) then it has a **Positive Skew (Right Skew)**.
- If the Median lies **closer to the Third Quartile** and if the **whisker at the upper end is shorter** then it has a **Negative Skew (Left Skew)**.

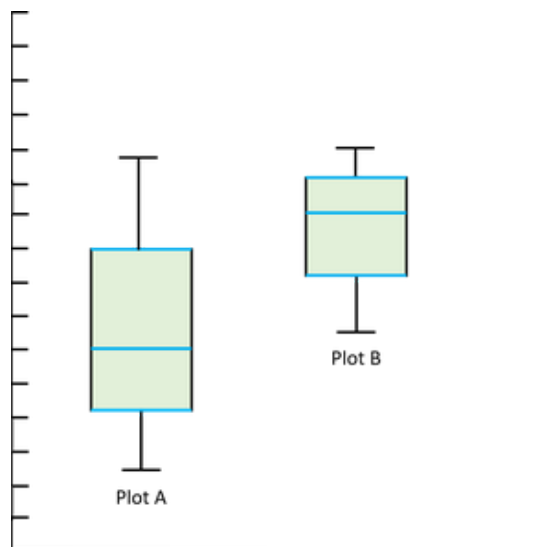


- The dispersion or spread of data can be visualized by the minimum and maximum values which are found at the end of the whiskers.
- The Box plot gives us the idea of about the Outliers which are the points which are numerically distant from the rest of the data.

4) How to compare box plots

As we have discussed at the beginning of the article that box plots make comparing characteristics of data between categories very easy. Let us have a look at how we can compare different box plots and derive statistical conclusions from them.

Let us take the below two plots as an example: -



- **Compare the Medians** — If the median line of a box plot lies outside the box of the other box plot with which it is being compared, then we can say that there is likely to be a difference between the two groups. Here the Median line of the plot B lies outside the box of Plot A.
- **Compare the Dispersion or Spread of data** — The Inter Quartile range (length of the box) gives us an idea about how dispersed the data is. Here Plot A has a longer length than Plot B which means that the dispersion of data is more in plot A as compared to plot B. The length of whiskers also gives an idea of the overall spread of data. The extreme values (minimum & maximum) gives the range of data distribution. Larger the range more scattered the data. Here Plot A has a larger range than Plot B.
- **Comparing Outliers** — The outliers gives the idea of unusual data values which are distant from the rest of the data. More number of Outliers means the prediction will be more uncertain. We can be more confident while predicting the values for a plot which has less or no outliers.
- **Compare Skewness** — Skewness gives us the direction and the magnitude of the lack of symmetry. We have discussed above how to identify skewness. Here Plot A is Positive or Right Skewed and Plot B is Negative or Left Skewed.

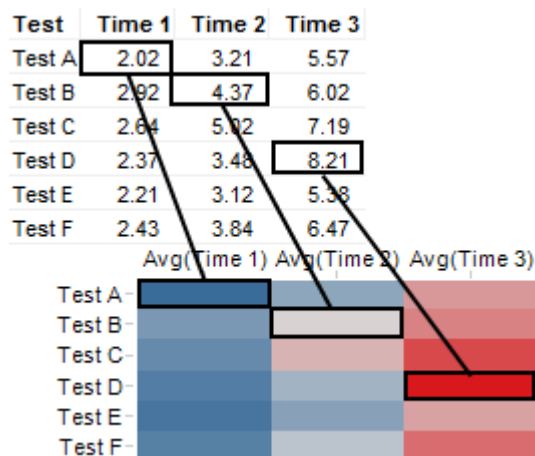
This is all for Box Plots. Now you might have got the idea of Box Plots how to make them and how to derive information from them. For any queries do leave a comment down below.

What is a Heat Map?

The easiest way to understand a heat map is to think of a cross table or spreadsheet which contains colors instead of numbers. The default color gradient sets the lowest value in the heat map to dark blue, the highest value to a bright red, and mid-range values to light gray, with a corresponding transition (or gradient) between these extremes. Heat maps are well-suited for visualizing large amounts of multi-dimensional data and can be used to identify clusters of rows with similar values, as these are displayed as areas of similar color.

Example:

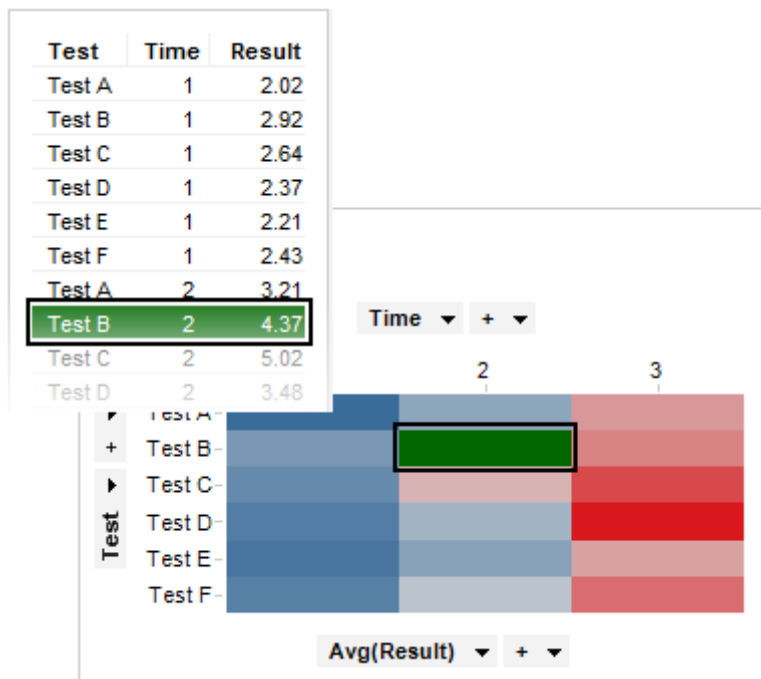
The example below shows how the values in the table are displayed as color gradients in the heat map cells.



As the examples below will illustrate; how to set up a heat map depends on the format of your data.

Data in tall/skinny format:

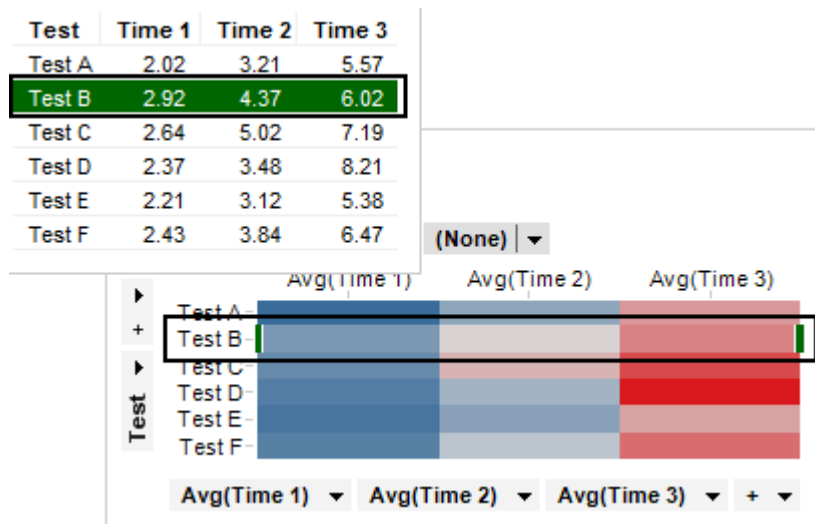
In this example, the data is in tall/skinny format, and each data table row corresponds to a single cell in the heat map.



The column Test is selected on the Y-axis, Time on the X-axis, and the cell values are set to the column Result. Like in other visualizations, highlighting and marking in the heat map are applied to one or more rows in the underlying data table. This means that with the setup in this example, marking a row in the table will mark a cell in the heat map.

Data in short/wide format:

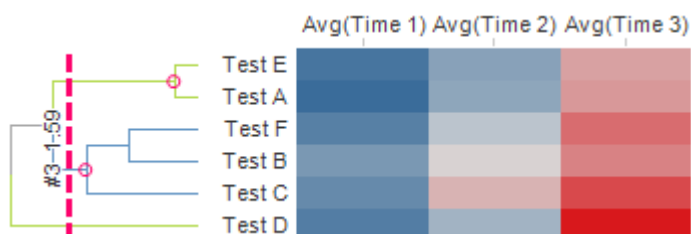
In this example, the data is in short/wide format instead, and each row in the data table corresponds to an entire row in the heat map.



The Y-axis in this heat map is set up with the column Test, while the X-axis is set to (None). For the individual cell values in the heat map, the columns Time 1, Time 2, and Time 3 are selected. Cell value columns are always aggregated unless the Y-axis is set to (Row Number). This is because the data table may contain many rows with the same name, and the values in these rows must then be aggregated to one single value to be displayed in the heat map. The content of the data is the same as in the example with data in tall/skinny format, but the format of the data makes it necessary to set up the heat map in a different way. With data in short/wide format, this is a common way to set up a heat map.

Dendrograms

It is often useful to combine heat maps with hierarchical clustering, which is a way of arranging items in a hierarchy based on the distance or similarity between them. The result of a clustering calculation is presented either as the distance or the similarity between the clustered items depending on the selected distance measure. See [Distance Measures Overview](#) and the detailed description for each measure for further information. To learn more about hierarchical clustering in general, see [Overview of Hierarchical Clustering Theory](#). You can cluster both rows and columns in the heat map. The result of a hierarchical clustering calculation is displayed in a heat map as a dendrogram, which is a tree-structure of the hierarchy. Row dendrograms show the distance (or similarity) between rows and which nodes each row belongs to as a result of the clustering calculation. Column dendrograms show the distance (or similarity) between the variables (the selected cell value columns). The example below shows a heat map with a row dendrogram where the distance between the rows were calculated.



As a result of the clustering calculation, the rows in the heat map have been reordered to correspond to the cluster calculation. Test A and Test E are placed in the same cluster. Test F and Test B are placed together in another cluster, and this cluster forms another cluster together with Test C. Test D is not included in any of those clusters. This indicates that Test A and Test E are closer to each other than what they are to Test F, Test B, Test

C, or Test D. It also indicates that Test D is the one that is the most distant to any of the other rows. See [Dendrograms and Clustering](#) to learn more about this.

All visualizations can be set up to show data limited by one or more markings in other visualizations only (details visualizations). Heat maps can also be limited by one or more filterings. Another alternative is to set up a heat map without any filtering at all. See [Limiting What is Shown in Visualizations](#) for more information.